bacteria, thus aiding in producing a clarified effluent. The succession of protozoa had long been observed in biological waste disposal systems.

Microbial waste management:

Generally, solid waste can broadly be categorized into biodegradable and non-biodegradable. The biodegradables (bio wastes) are those solid wastes generated, which could be decomposed by micro organisms and does not constitute major sources of pollution for a long period of time. They include paper products and wastes of plant origin, wastes of animal origin (faecal matter, carcass, droppings, and poultry waste products). These groups of solid waste even though they are easily degraded by microorganism in minimal time, give off offensive odour and constitute nuisance to the aesthetic environment more than the non-biodegradable solid wastes. On the other hand, non-biodegradable solid wastes are not degradable by microorganisms. This implies that other means of treatment such as incineration, landfill, and recycling are employed as ways of disposing them.

Municipal solid waste management:

Municipal solid waste management reduces or lessens its effect on the mankind and leads to increase in economic development of country and maintain the sustenance of life. Composting is used as biological solid waste treatment method which is the controlled aerobic decomposition of organic waste materials with the help of microorganisms. Composting is a method in which organic waste matte are decomposed and then recycled as compost for use in agriculture and landscaping applications. The most common composting techniques include static pile composting& vermin-composting.

Recent advances in microbial waste management:

The review of recent researcher activity which is applied in microbes to both environmental management and biotechniques is informed by the polluting effects on the world by soil erosion, harmful fertilizers and pesticides, and the improper treatment of sewage these harmful processes have caused in serious environmental and social problems around the world. Notably, biological methods are being used in sustainable environmental and waste management techniques. Some of the techniques are bioremediation, biostimulation, bioaugmentation, phytoremediation, and so on.

CONCLUSION

Waste is any material, which is of no use to anyone. Humans with nearly all activities produce waste. The major component of municipal solid waste is organic fraction, mostly from domestic, agricultural and industrial sources. There are many different methods of municipal waste management. These include physical, chemical and biological methods. Conventional waste management practices usually involve one negative consequence or the other. This necessitated the search for and development of biological methods, with the use of microorganisms which produce environmental-friendly results. Biological methods include the use of microorganisms such as bacteria, fungi, algae, virus and protozoa in process like composting, activated sludge, trickling filters and oxidation ponds. Recent scientific progress in applying microorganisms to environmental management includes hybrid techniques that combine microbial methods with physical and chemical ones; these include landfills, anaerobic digestion and vermiculture technology.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest.

REFERENCES

1. **Brewer, R. and Belzer, W., 2001.** Assessment of metal concentrations in atmospheric particles from Burnaby Lake, British Columbia, Canada. Atmospheric Environment. **35(30):** 5223-5233.
2. **Liu, G., Zhu, Z., Yang, Y., Sun, Y., Yu, F. and Ma, J., 2019.** Sorption behavior and mechanism of hydrophilic organic chemicals to virgin and aged microplastics in freshwater and seawater. *Environmental Pollution*, **246:** 26-33.
3. **CPCB. 2000.** Air quality status and trend in India. Parivesh Newsletter, vol. **4(3)**. Central Pollution Control Board, Parivesh Bhavan, Delhi, India.
4. **Karthick, T., Sumithra, B., Bhuvaneshwari, K.P., Gurujeya, B. and Jaya Barath, J., 2016.** Biodegradation of plastic by *Bacillus subtilis* isolated from polythene polluted soil. *World J. Pharmaceut. Res*. **5:** 1148-1153.

ADDITIONAL REFERENCES

1. **Abila, B., and Kantola, J. 2013.** Municipal waste management problems in Nigeria: Evolving knowledge management solution. *International Journal of Environmental, Ecological, Geological and Mining Engineering*. **7(6):** 172-177.

2. **Acharya, A., Joshi, D.R., Shrestha, K. and Bhatta, D.R., 2012.** Isolation and screening of thermophilic cellulolytic bacteria from compost piles. *Scientific world*. **10(10):**43-46.

3. **Al-Salem, S.M., Evangelisti, S. and Lettieri, P., 2014.** Life cycle assessment of alternative technologies for municipal solid waste and plastic solid waste management in the Greater London area. *Chemical Engineering Journal*. **244:**391-402.

4. **Hussein, A.A., Al-Mayaly, I.K. and Khudeir, S.H., 2015.** Isolation, Screening and Identification of Low Density Polyethylene (LDPE) degrading bacteria from contaminated soil with plastic wastes. *Mesopotamia Environ. J.* **1(4):**1-14.

5. **Amalesh Samanta, Debmalya Mitra, Sudipendra Nath Roy, Chandrima Sinha, Pinaki Pal. 2013.** Characterization and Optimization of Amylase Producing Bacteria Isolated from Solid Waste. *Journal of Environmental Protection.* **4(6):** 647-652.

6. **Amrita Saha and Subhas Chandra Santra. 2014.** Isolation and characterization of bacteria isolated from municipal solid waste for production of industrial Enzymes and Waste Degradation. *Journal of Microbiology & Experimentation*. **1(1):**1-8.

***

Biospectra : ISSN: 0973-7057, Vol. 17(1), March, 2022, pp. 55-58 **11(PS).**
An International Biannual Refereed Journal of Life Sciences **Plant Science**

ISSN : 0973-7057

Int. Database Index: 616 www.mjl.clarivate.com

Proximate composition of some underutilized edible wild plants mainly consumed by Munda tribe of Jharkhand

Shalini Kumari* & Jyoti Kumar
University Department of Botany, Ranchi University, Ranchi, Jharkhand, India

Received : 15th November, 2021 ; Revised : 4th January, 2022

Abstract-Underutilized edible wild herbaceous plants mainly consumed by Munda tribes of Jharkhand namely *Senna tora, Commelina benghalensis, Alternanthera paronychioides* were analyzed for their proximate composition. Proximate composition was determined by using standard methods of Association of Official Analytical Chemists (AOAC). The result showed that proximate composition of the selected wild edibles ranged between; moisture (9.56±0.2 - 11.15±0.13%), ash content (9.82±0.12 - 11.56±0.14%), crude protein (6.12± 0.11 - 6.52±0.18%), crude fiber (4.78±0.1 - 6.15±0.2%), fat (1.16±0.08 - 1.91±0.11%), carbohydrate (69.33±0.58 - 73.06±0.76%), caloric value (317.1±0.61 -328.38±1.2 kcal/100g). The varied proportion of nutrient is found in this wild edibles which helps to obtain a scientific data baseline for their wider acceptance as additional, nutritional supplementary source of food.

Key words: Underutilized edible wild, Proximate composition, Munda tribe, Jharkhand

INTRODUCTION

Jharkhand (The land of forest) is primarily known as tribal state with total tribal population of 70,89,068 which is about 28% of state population. There are 32 tribal communities inhabiting here in which Munda, Oraon, Ho, Ashur are in majority. In Jharkhand most of the rural inhabitants depend on wild edible plants to meet their additional food requirements. The wild edible plants have traditionally occupied an important position in the socio-cultural, spiritual and health arena of rural and tribal lives in Jharkhand.

The wild plants from forest provides many essential nutrients which helps to improve both the physical and mental well-being of rural people. In developing countries, the wild plants have been a main source of food and medicine for tribal people. It is estimated that 80% of the forest dwellers depend on forests for 25-50% of their annual food requirements.[1] They are inexpensive, easy to cook and rich sources of macro and micro nutrients.[2,3]

Proximate and nutritional analysis of edible fruit and vegetables play a crucial role in assessing their nutritional significance.[4] Since the selected wild edible herbaceous plants are underutilized, information on their nutritional and medicinal properties are lacking. As there are not many published studies on nutritional and chemical composition of these wild edible varieties, so there is need to initiate the awareness for its wider acceptability among mass urban population. In this research work underutilized edible wild herbaceous plants used by Munda tribe namely *Senna tora, Commelina benghalensis and Alternanthera*

*Corresponding author :
Phone : 9073619428
E-mail : keshrishalu177@gmail.com

paronychioides were analyzed for proximate composition using standard AOAC method.

MATERIALS & METHODS

Sample collection and preparation:

Four underutilized edible wild herbaceous plants used by Munda tribe namely *Senna tora, Commelina benghalensis and Alternanthera paronychioides* were collected from different localities of Ranchi (Jharkhand).

After collection, the leaves were removed from the branches and were sorted out to get the healthy and disease free edible parts. The leaves were washed properly and dried them under shade at 25 $\pm$ 2°C so as to prevent the decomposition of chemical compounds present in them. The dried plant sample were powdered by just crushing them, sieved through 2 mm mesh sieve and stored in air tight container at room temperature for further investigation.

Methods

Nutritional composition of these dried leafy plant samples were analyzed by using standard method of Indian Standard and Association of Official Analytical Chemists (AOAC)[5] methods. The moisture content was determined by oven dehydration at 103°C for 6 hours. The ash content of the sample was determined by keeping the sample in silica crucible dish and ignited them in muffle furnace at 525 $\pm$ 25 °C until greyish white ash results, then it is cooled in a desiccators and weighed the incinerated residue, according to Association of Official Analytical Chemists (AOAC)[5] method. The total protein content was estimated by Micro-Kjeldahl method.[5]

Crude fat or oil were extracted by Soxhlet extraction with petroleum ether for 16h.[5] Crude fibre were determined by acid- base digestion method.[6] Total carbohydrate content was estimated by difference method as described by James (1995)[7]. Add moisture, fat, protein and ash content and deduct the value from 100 to get carbohydrate content by difference. Caloric value (Kcal/ 100g) were estimated by using the method described by FAO (2003)[8] i.e., {4× (% of protein)} + {9×(% of fat)} + {4×(% of carbohydrate)}.

RESULT & DISCUSSION

Some selected species of underutilized edible wild plant showed variant proportions of proximate composition as mentioned in the Table 1.

Moisture Content

Moisture in vegetables is a good source of water and is necessary as it is considered that around 20% of the total water consumption must come from food moisture.[9] On dry basis the moisture content of the underutilized edible wild species *Senna tora, Commelina benghalensis* and *Alternanthera paronychioides* were 9.56 %, 10.29 % and 11.18 % respectively. The value of moisture content was comparable to other commonly cultivated species such as *Spinacia oleracea, Coriandrum sativum* and *Brassica oleraceae* were 14.51 %, 10.49 % and 24.00 % respectively.[10]

Ash Content

From the biochemical point of view, ash content plays an important role as it contains important nutritional elements mainly minerals which is helpful in the physiological function of the body. The ash content of the underutilized edible wild herbaceous species *Senna tora, Commelina benghalensis* and *Alternanthera paronychioides* were 8.82%, 13.85% and 16.84 % respectively. The ash value were more or less same, compared with the commonly cultivated species such as *Spinacia oleracea, Coriandrum sativum* and *Brassica oleraceae* were 22.50 %, 20.07% & 8.21% respectively.[10]

Crude Fat Content

Fats are a type of lipids which is stored in the adipose tissue and under the skin of animals. It is mainly used as an energy storage molecule in the body as 1gram fat gives 9 Kcal energy. The fat content of the selected edible wild species *Senna tora, Commelina benghalensis* and *Alternanthera paronychioides* were 1.18%, 1.93% and 1.18% respectively. The crude fat content of the selected wild species were lower as compared to 7.12% in *Spinacia oleraceae*[11] and are same when compared with other wild edibles like *Ipomoea aquatica* (1.2%), *Lippia alba* (1.39%), *Cayratia trifolia* (1.61%)[12]. Due to the general low level of crude fat in vegetable leaves and its high level of total unsaturated fatty acid, their consumption in large amount would be beneficial to individual suffering from overweight or obesity and this would constitute a good dietary habits.

Crude Fibre Content

Fibre is the indigestible part of plant food which is important for our digestive health. It can improve cholesterol and blood sugar level and helps in preventing